

1M1B • IBM SKILLSBUILD • AICTE INTERNSHIP

EcoSort AI

Smart Waste Segregation & Circular Recovery Assistant

AI-powered source-level waste classification guiding users to achieve UN SDG 12 targets.

SUBMITTED BY

Pawan Tiwari

B.Tech Computer Science & Engineering
Ambalika Institute of Management & Technology
AICTE STU ID: STU6a4f981e559f11783601182

PROJECT METRICS

- Primary SDG: SDG 12 (Target 12.5)
- Core Engine: IBM Granite / LLM Logic
- Interface: Reactive Streamlit UI
- Mode: Real-time Multi-factor Triage

SDG Alignment & The Ground Reality Problem

Overcoming contamination at source through intelligent triage.

THE PROBLEM STATEMENT

Commingled Waste Streams

- Source Mixing: Wet organic food, dry recyclables, and toxic cells are dumped into single bins.
- Landfill Burden: Over 70% of recyclables get ruined and degrade into methane in open dumps.
- Occupational Risk: Exposes sanitation workers to sharp glassware and chemical battery leaks.

SDG ALIGNMENT

UN Sustainable Goals

- Primary: SDG 12 (Responsible Consumption & Production - Target 12.5).
- Secondary: SDG 11 (Sustainable Cities and Communities - Target 11.6).
- Objective: Preventing recyclables from ending up in open dumps through source-level AI precision.

HOW MIGHT WE (HMW) STATEMENT

"How might we use AI to guide users in segregating waste at the exact point of disposal so that waste segregation and material recovery can become accurate, scalable, and habitual?"

Target Users & The Need for AI

Bridging the gap between static signage and active segregation habits.

WHO BENEFITS

Target Users

- Students & campus faculties
- Cafeteria & food court staff
- Residential housing societies
- Municipal sanitation teams

WHY POSTERS FAIL

Static Poster Limits

- Ignored passive signage
- No help on composite items
- Zero pre-disposal guidance
- No habit reinforcement loop

HOW AI TRANSFORMS

The AI Advantage

- Instant real-time precision
- Multimodal text/image triage
- Prep guidance (rinse/crush)
- Measurable sustainability tips

AI Solution Architecture & Data Flow

From user query to color-coded municipal action.

01

USER INPUT

Query / Capture

User enters item name (e.g. 'banana peel') or submits an image frame.

02

AI ENGINE

Granite / LLM

Extracts material profile: Organic pulp, PET #1, laminates, or heavy metals.

03

TRIAGE RULES

Municipal Logic

Maps contamination level against standard color bins (Green, Blue, Red/Black).

04

OUTPUT

Actionable Output

Delivers exact bin, mandatory preparation steps, and environmental benefits.

Responsible AI Considerations

Aligned strictly with IBM SkillsBuild ethical AI frameworks.

INCLUSIVE MODEL

Fairness & Localization

Calibrated for Indian and regional waste types (tea kulhads, coconut husks, snack pouches) to prevent classification bias.

HABIT FORMATION

Explainability

Explains the 'Why' behind bin allocation so users learn the mechanics of recycling instead of relying on a black box.

ZERO BIOMETRICS

Privacy by Design

Operates exclusively on object data. Zero facial imagery, personal telemetry, or identifiable data is captured or saved.

PHYSICAL SAFETY

Safety & Hazard Flags

Detects toxic items (lithium batteries, broken bulbs, solvents) and triggers bold visual safety warnings instantly.

Prototype Demonstration & Test Cases

Verified interactive Streamlit assistant in live operation.

ACTIVE STREAMLIT GUI

Live Execution Result

VALIDATED TEST CASES

Inference Verification

 Banana Peel

 **Green Bin (Wet/Compost)**

Discard directly without plastic bag.

 Plastic Bottle

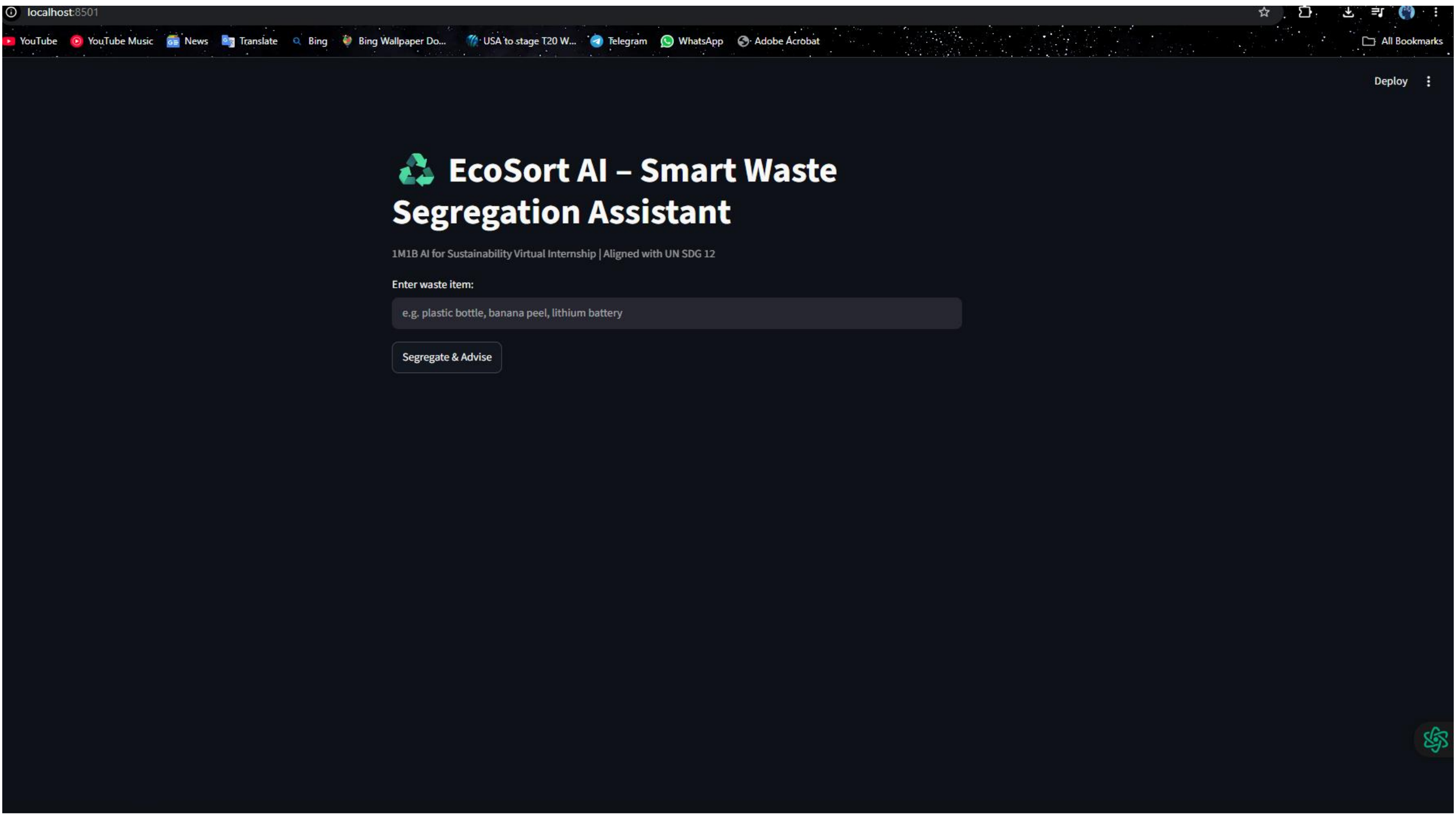
 **Blue Bin (Dry Recyclable)**

Rinse residual liquid and crush body.

 Lithium Battery

 **Red/Black Bin (Hazardous)**

Insulate terminals with tape before dropoff.



EcoSort AI – Smart Waste Segregation Assistant

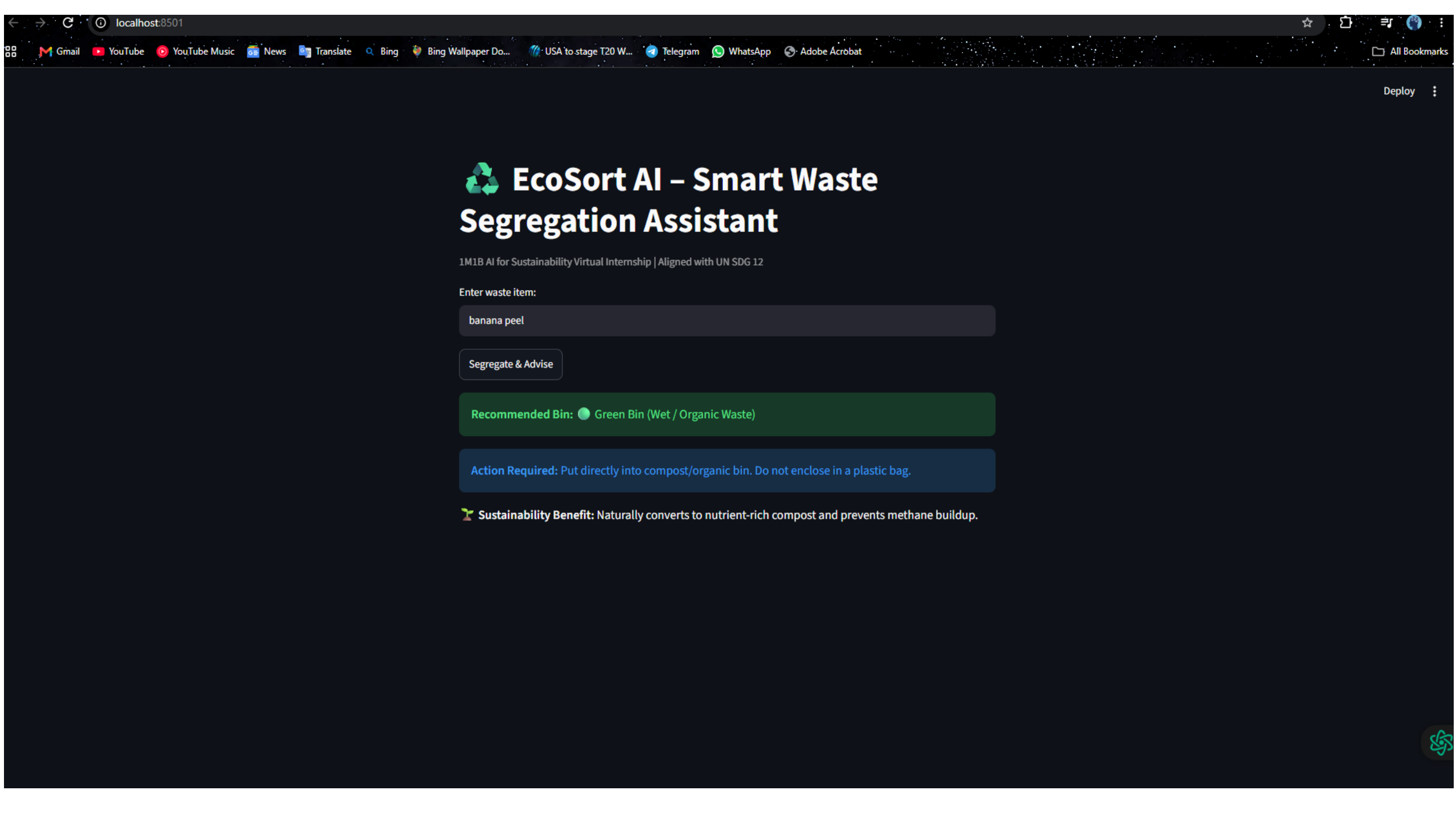
1M1B AI for Sustainability Virtual Internship | Aligned with UN SDG 12

Enter waste item:

e.g. plastic bottle, banana peel, lithium battery

Segregate & Advise





EcoSort AI – Smart Waste Segregation Assistant

1M1B AI for Sustainability Virtual Internship | Aligned with UN SDG 12

Enter waste item:

banana peel

Segregate & Advise

Recommended Bin: ● Green Bin (Wet / Organic Waste)

Action Required: Put directly into compost/organic bin. Do not enclose in a plastic bag.



Sustainability Benefit: Naturally converts to nutrient-rich compost and prevents methane buildup.



Expected Impact & Future Scope

Quantifiable environmental progress and technological roadmap.

PLANET

Environmental Impact

- Significant reduction in landfill burden
- Eliminates anaerobic methane production
- Protects recyclables from wet damage
- Maximizes circular plastic recovery

PEOPLE

Worker Dignity

- Protects waste staff from sharp cuts
- Eliminates toxic chemical contact
- Lowers manual segregation burden
- Instills responsible campus civic habits

ROADMAP

Campus Metric & Future

- Helps campus meet Green Audit scores
- Swachh Campus ranking benchmark
- IoT smart dustbins with camera lids
- Automatic mechanical sorting flap triggers